Mayank Manohar Polepalli

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Software Engineer

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About: Early-career software engineer with strong computer science fundamentals and applied data structures and algorithms in distributed systems and backend work. Proficient in C/C++, Java, Go, and Python; ships production features behind RESTful APIs and microservices architecture, with Generative AI integration experience and AI-assisted development tooling including Cursor and Claude

Experience

Software Developer Trainee (Full Time) **Sep 2025 — Sep 2026**
Manhattan Associates Bengaluru

- Implemented RESTful APIs and microservices architecture with optimized database queries, caching strategies, and disciplined automated testing
- Developed and maintained web applications using Angular, Spring Boot, and MySQL in Agile delivery with CI/CD pipeline contributions
- Collaborated with cross-functional engineering teams on design, code review, and production reliability

Teaching Assistant (Part-Time) **Aug 2024 — Dec 2024**
PES University Bengaluru

- Conducted 20+ lab sessions for Database Management Systems, assisting 50+ students with SQL debugging, workflow validation, and data integrity checks
- Built Python-based automated checks for student submissions, reducing manual grading effort and improving evaluation consistency

Projects

Distributed Key-Value Store **Oct 2025**

- Built a distributed key-value store in Go implementing the Raft consensus algorithm with leader election, heartbeat-based failure detection, and append-only log replication across nodes
- Designed an LSM-tree storage engine (memtable, sorted string tables, bloom filters) with consistent hashing for partition assignment and validated linearizability under node crashes and network partitions

Hybrid L4/L7 Load Balancer **Nov 2025**

- Implemented a concurrent load balancer in Go handling Layer 4 (raw TCP proxy) and Layer 7 (HTTP routing with header inspection) traffic, supporting round-robin, least-connections, and weighted scheduling
- Built connection pooling with keep-alive management, active/passive health checks, and automatic failover; validated correctness with fault injection, synthetic traffic generation, and race-condition testing

Research & Publications

Research Assistant, IISc Bengaluru **Sep 2025 — Mar 2026**

- Developed a Generative AI multi-agent orchestration framework (LangGraph, FastAPI, Redis) with automated test pipelines and runtime state monitoring

ICICIP 2025 Conference, Oman (Accepted) **2025**

- Published ML research on remote sensing data for migratory bird-strike risk prediction at airports, with emphasis on model validation and reliability

Education

Master of Science, Computer Science - University of Massachusetts Amherst **Sep 2026 — Present**
Bachelor of Technology, CSE - PES University, 8.12 CGPA **Sep 2021 — Aug 2025**

Prior: PUC 94% (RVPU College); CBSE 93% (Sri Kumaran Children's Academy); ICSE (Clarence Public School)

Skills

Programming Languages	C/C++, Java, Go, Python, JavaScript, TypeScript, Unix Shell,
CS foundations	Data structures & algorithms, distributed systems, concurrency and networking (applied in systems projects),
Backend & data	Spring Boot, FastAPI, REST APIs, microservices architecture, MySQL, PostgreSQL, MongoDB, Redis,
GenAI & AI tooling	Generative AI engineering, Langchain, PyTorch, vLLM, ChatGPT, Claude Code, Cursor, Gemini API,
Quality & DevOps	CI/CD pipelines, Playwright, Selenium, JUnit, TestNG, Jenkins, SonarQube, GitHub Actions, GitLab CI, Docker, Grafana, Git, JIRA,